

DECKBench: Benchmarking Multi-Agent Frameworks for Academic Slide Generation and Editing

Verified reference summary • Year: 2026 • Source bibliography: docs/REFERENCES.md

What the paper contributes

DECKBench focuses on multi-agent slide generation and editing, including slide-level and deck-level fidelity, coherence, layout quality, and multi-turn instruction following.

Method / evaluation stance

The paper is relevant because slidekit is explicitly agent-first. Multi-agent workflows need robust handoffs: one agent may define brand, another outline content, and another build the deck. Free-form HTML can make those handoffs brittle.

How slidekit uses it

YAML offers a compact contract. A brand agent can produce a theme block; a storyteller can produce slide intents; a deck builder can select components and fill slots. Validation then catches missing or malformed fields.

YAML vs HTML takeaway

HTML remains useful when an agent must build a highly custom visual surface. For the repeatable deck-generation workflow in this repo, constrained YAML reduces coordination cost and makes edits easier to review.

Bottom line

The paper supports a pragmatic split: use research-style rendered artifacts when the question depends on final pixels or human preference collection, but keep slidekit production decks in a constrained YAML IR so layout, linting, and agent fixes remain deterministic, inspectable, and cheap.

Primary slidekit mapping	balance / alignment / whitespace / overlap / contrast / richness / consistency, depending on the paper
Operational risk	HTML metrics can drift with rendering environment; YAML metrics can miss pixel-only effects. Use each where it is strongest.